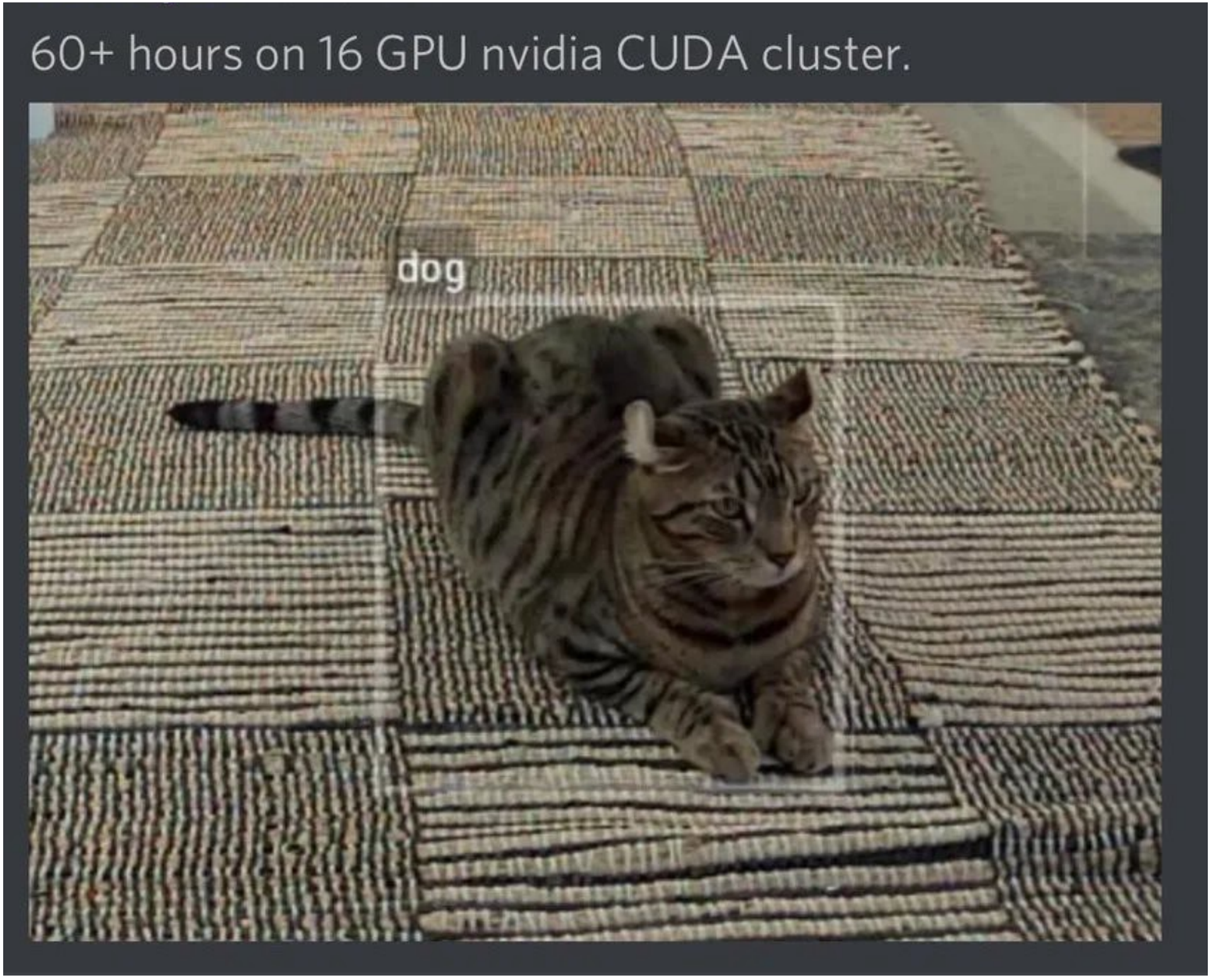
Clustering Analysis and K-Means

Mahdi Roozbahani
Georgia Tech

Some of the slides are inspired based on slides from Chao Zhang, and Le Song.


60+ hours on 16 GPU nvidia CUDA cluster.
dog

$X = \begin{bmatrix} & \\ & \end{bmatrix}_{n \times d}$ $Y = \begin{bmatrix} \\ \end{bmatrix}_{n \times 1}$

unsupervised learning ⇝ we only have access to X ⇝ clustering

⇝ ML

Supervised learning ⇝ we have access to both X and Y

Regression

↳ Y ⇝ continous

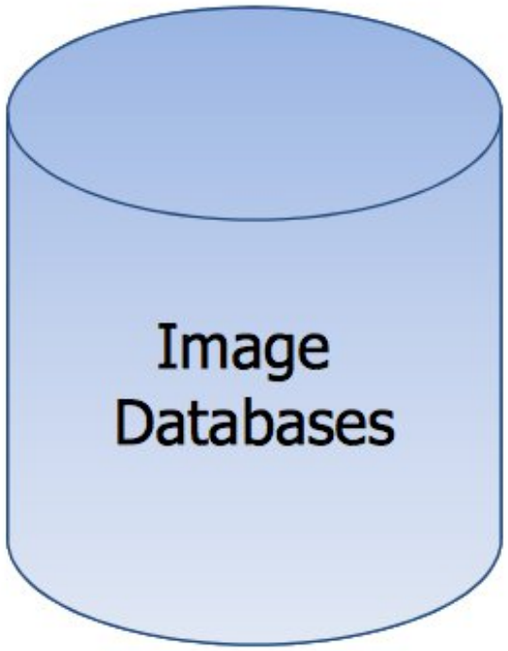
Classification

↳ Y ⇝ Discrete

Outline

- Clustering
- Distance Function
- K-Means Algorithm
- Analysis of K-Means

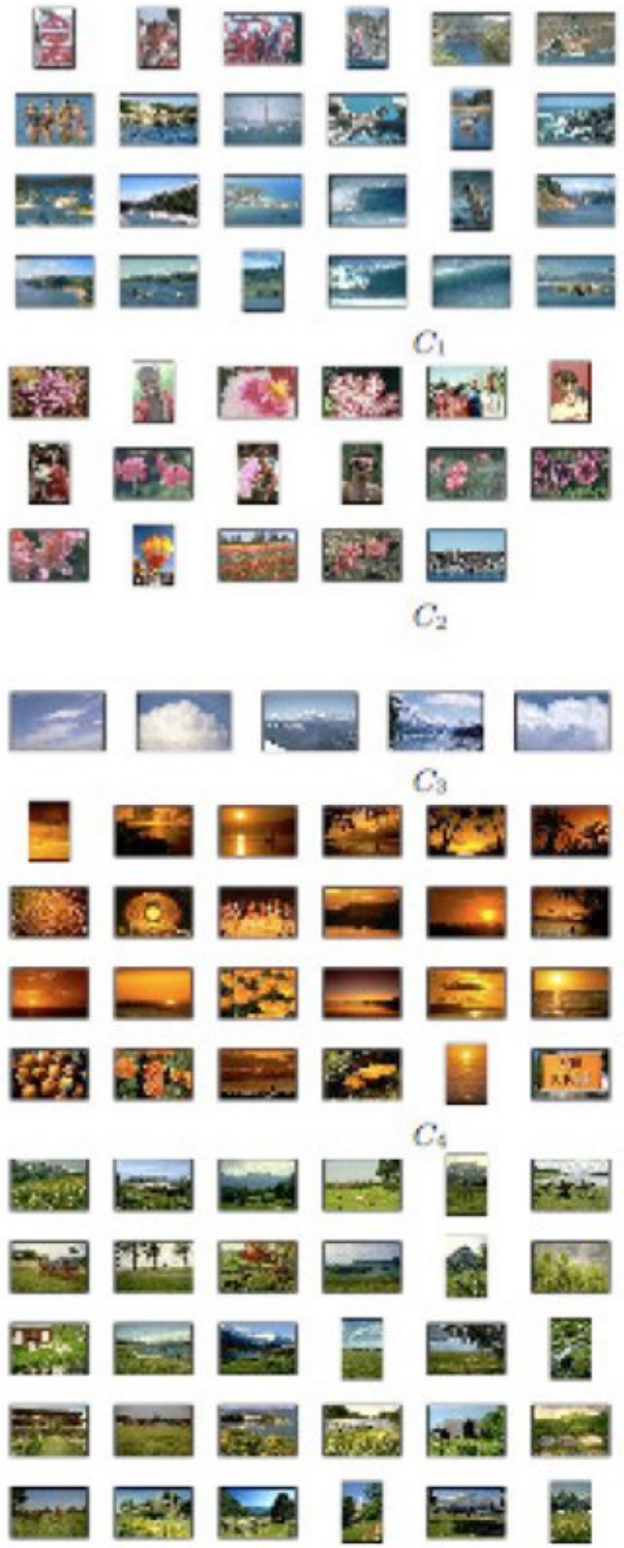
Clustering Images



C ⇝ cluster

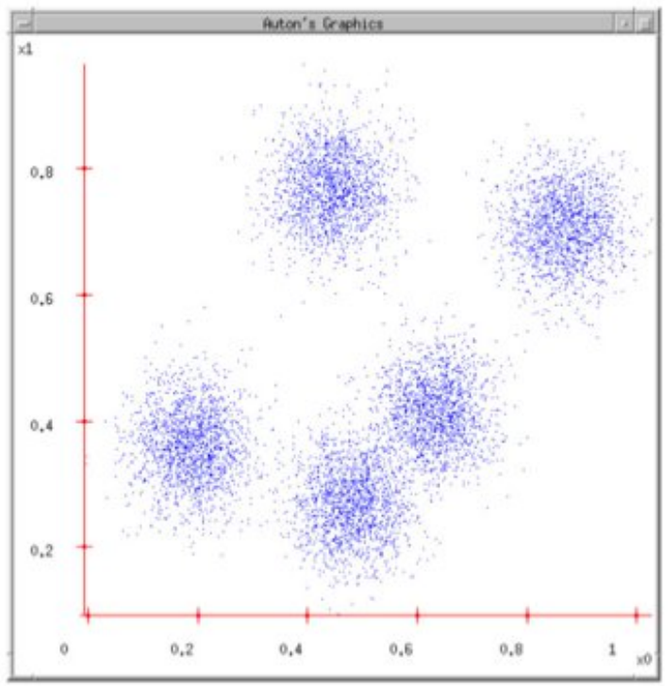
k ⇝ # of clusters

k=5

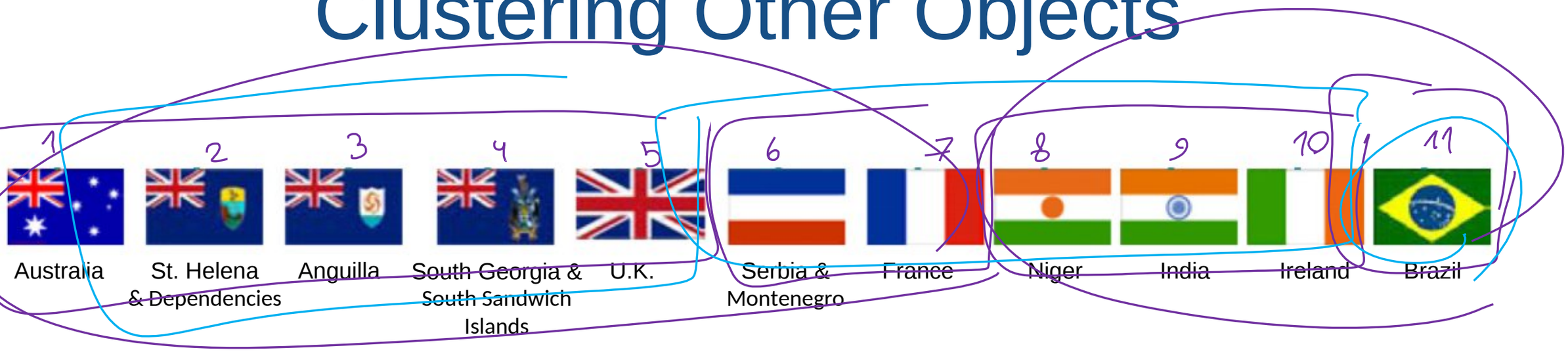


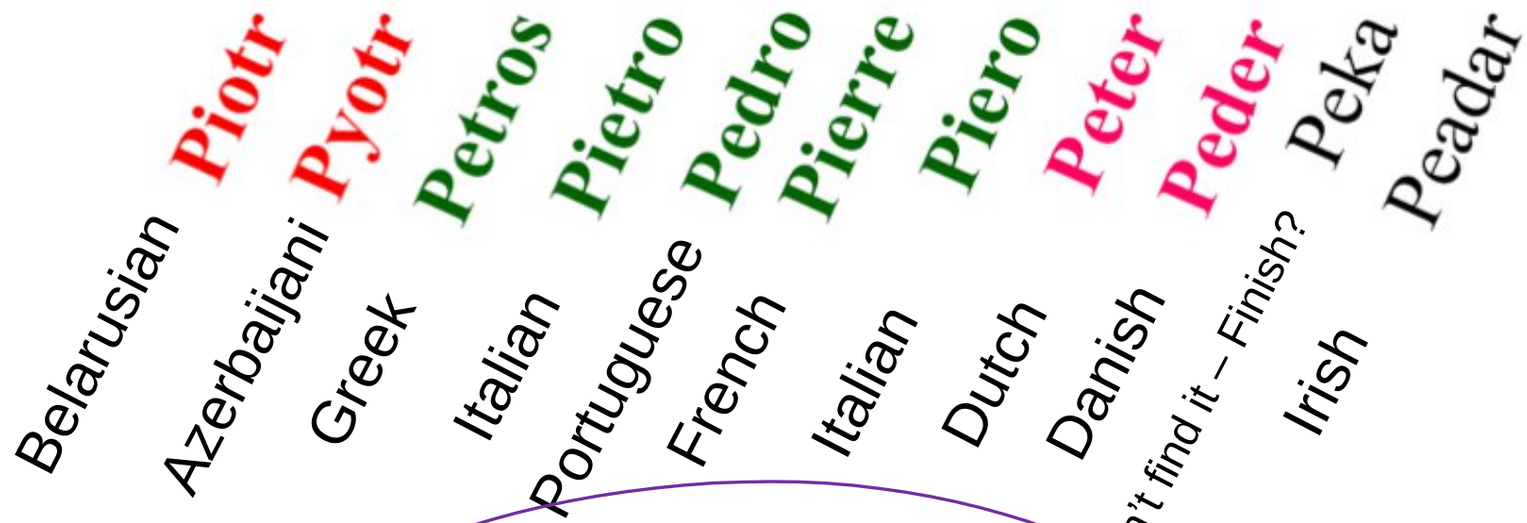
Goal of clustering:

Divide object into groups, and objects within a group are more similar than those outside the group

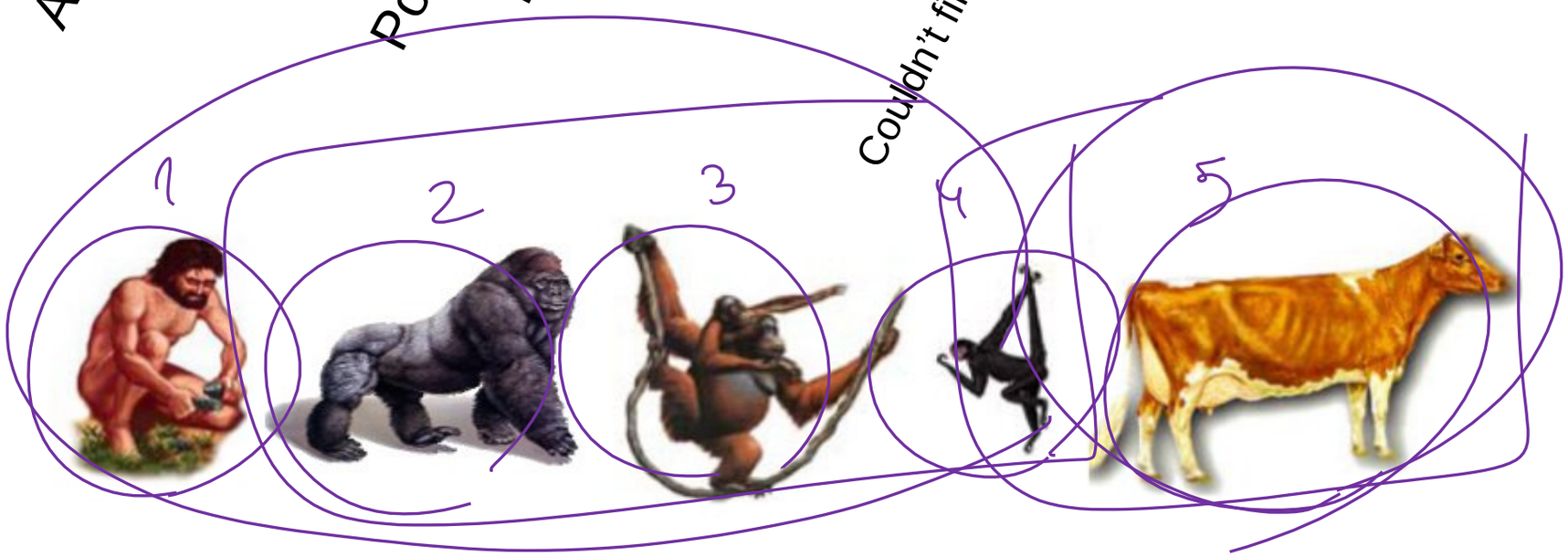


Clustering Other Objects



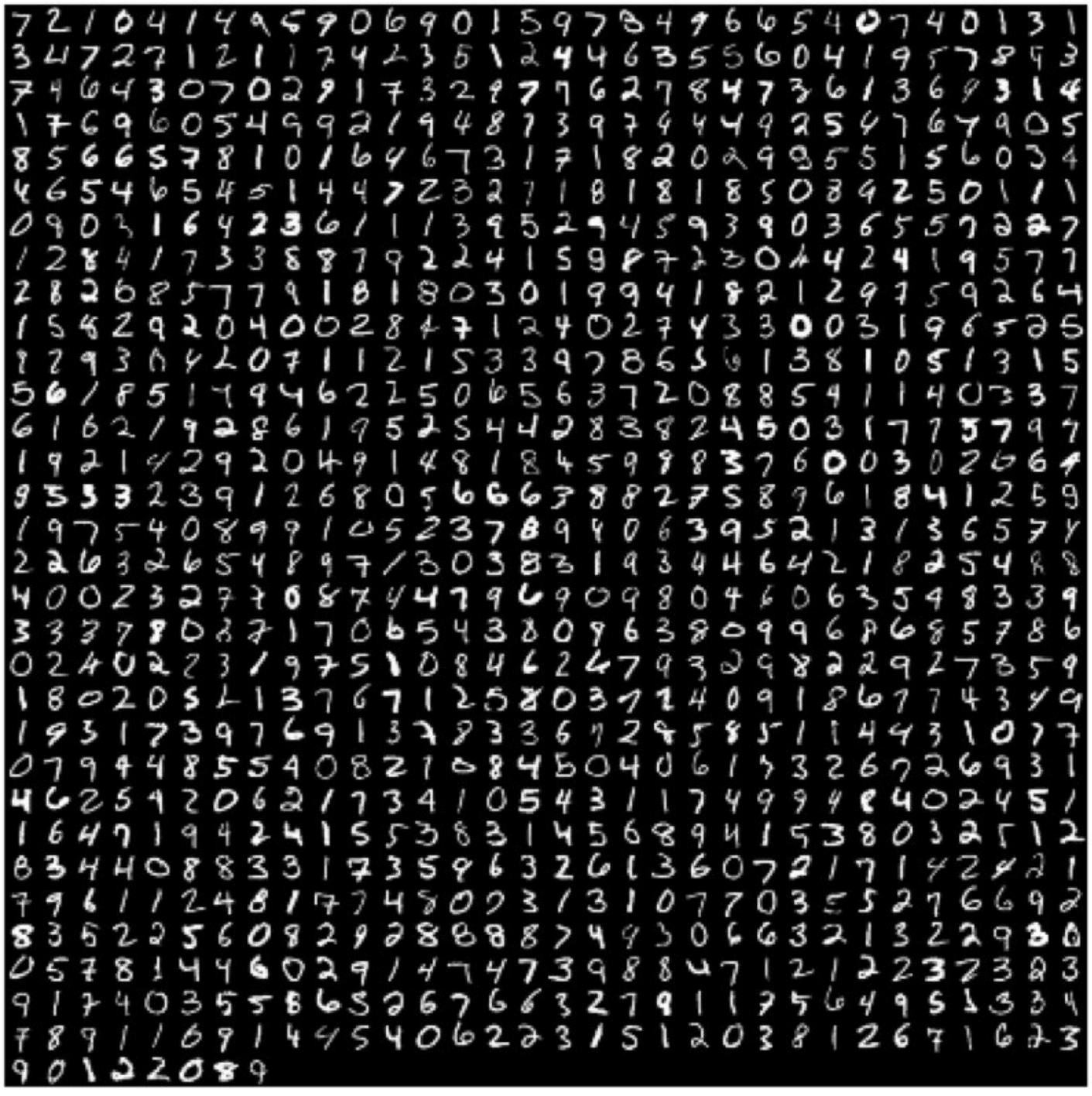


Linguistic Similarity

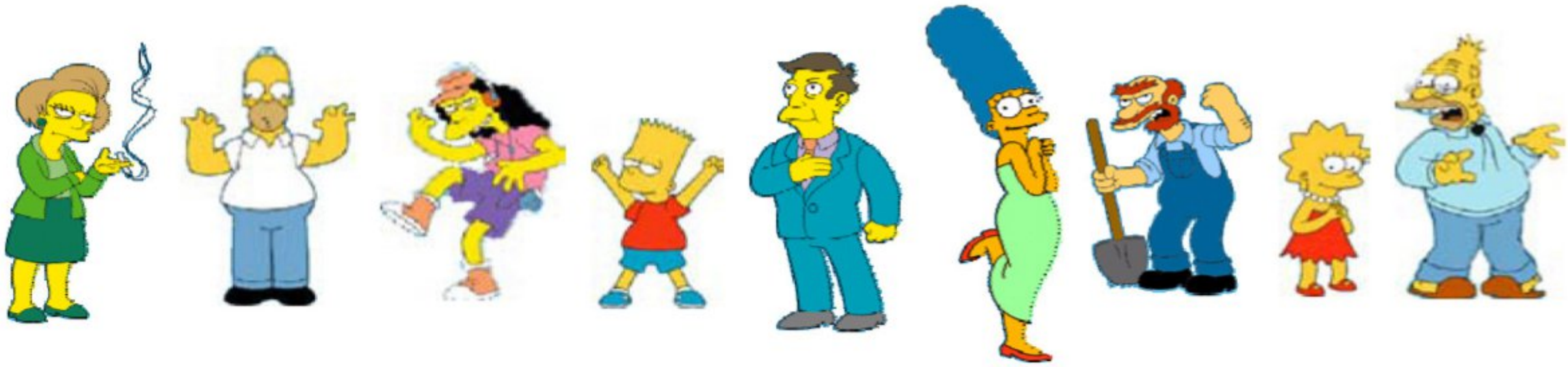


Clustering Hand Digits

0 1 2 3 4 5 6 7 8 9

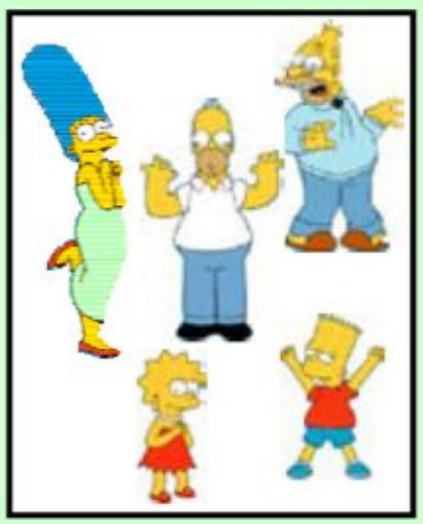


Clustering is Subjective



What is consider similar/dissimilar?

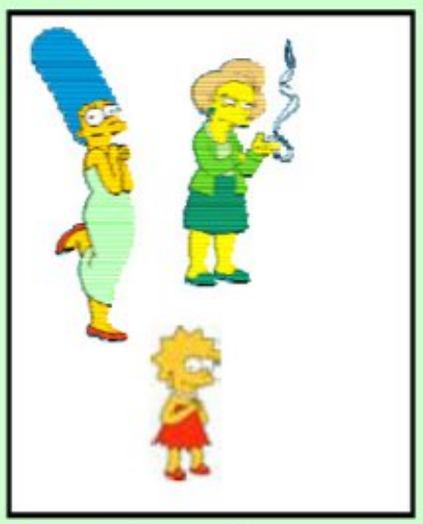
Clustering is subjective



Simpson's Family



School Employees



Females



Males

Are they similar or not?

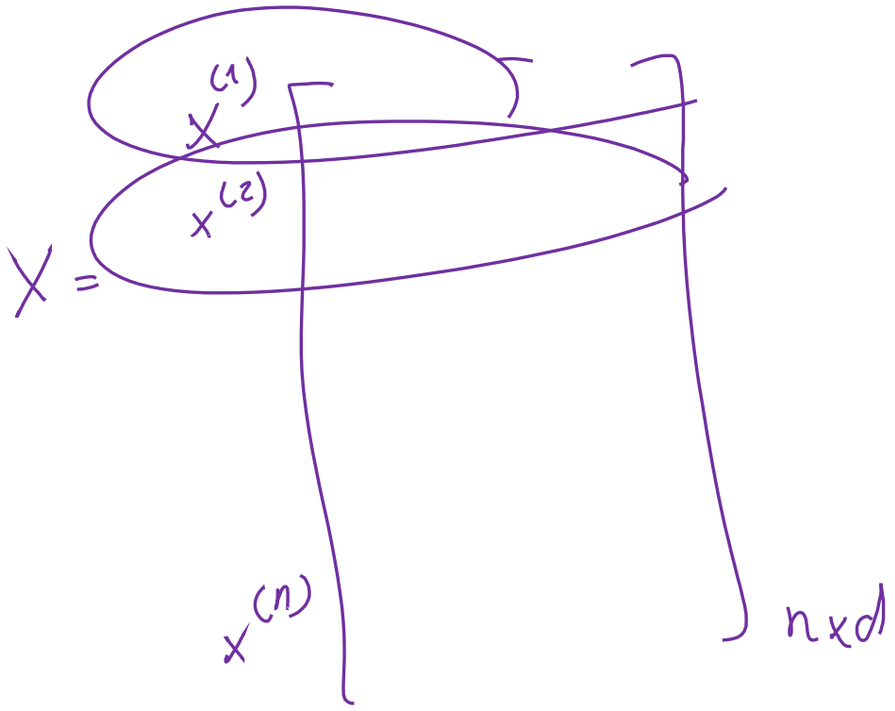


So What is Clustering in General?

- You pick your similarity/dissimilarity function
- The algorithm figures out the grouping of objects based on the chosen similarity/dissimilarity function
  - Points within a cluster is similar
  - Points across clusters are not so similar
- Issues for clustering
  - How to represent objects? (Vector space? Normalization?)
  - What is a similarity/dissimilarity function for your data?
  - What are the algorithm steps?

Outline

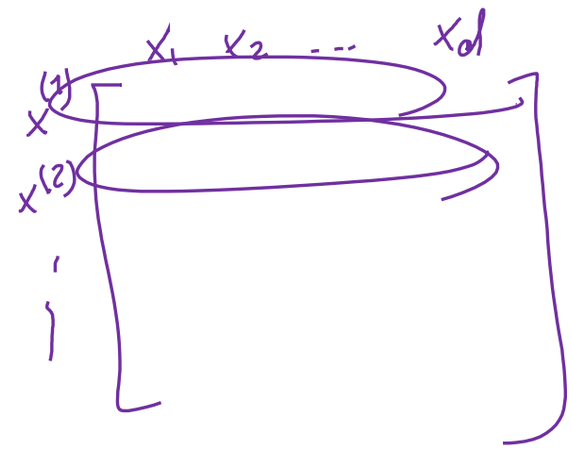
- Clustering
- Distance Function
- K-Means Algorithm
- Analysis of K-Means



Properties of Similarity Function

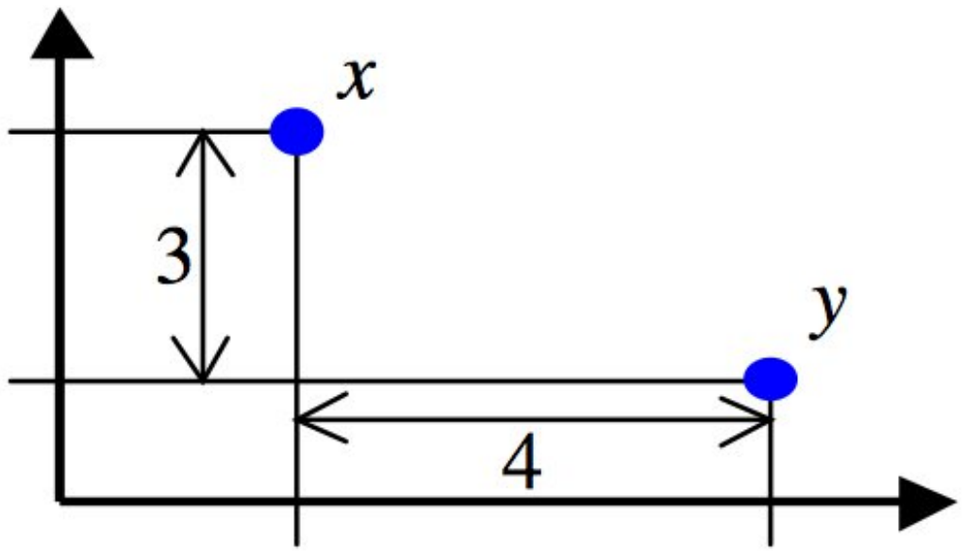
- Desired properties of dissimilarity function
  - Symmetry: $d(x, y) = d(y, x)$
    - *Otherwise you could claim "Alex looks like Bob, but Bob looks nothing like Alex“*
  - Positive separability: $d(x, y) = 0$, if and only if $x = y$
    - *Otherwise there are objects that are different, but you cannot tell apart*
  - Triangular inequality: $d(x, y) \leq d(x, z) + d(z, y)$
    - *Otherwise you could claim "Alex is very like Bob, and Alex is very like Carl, but Bob is very unlike Carl"*

Distance Functions for Vectors



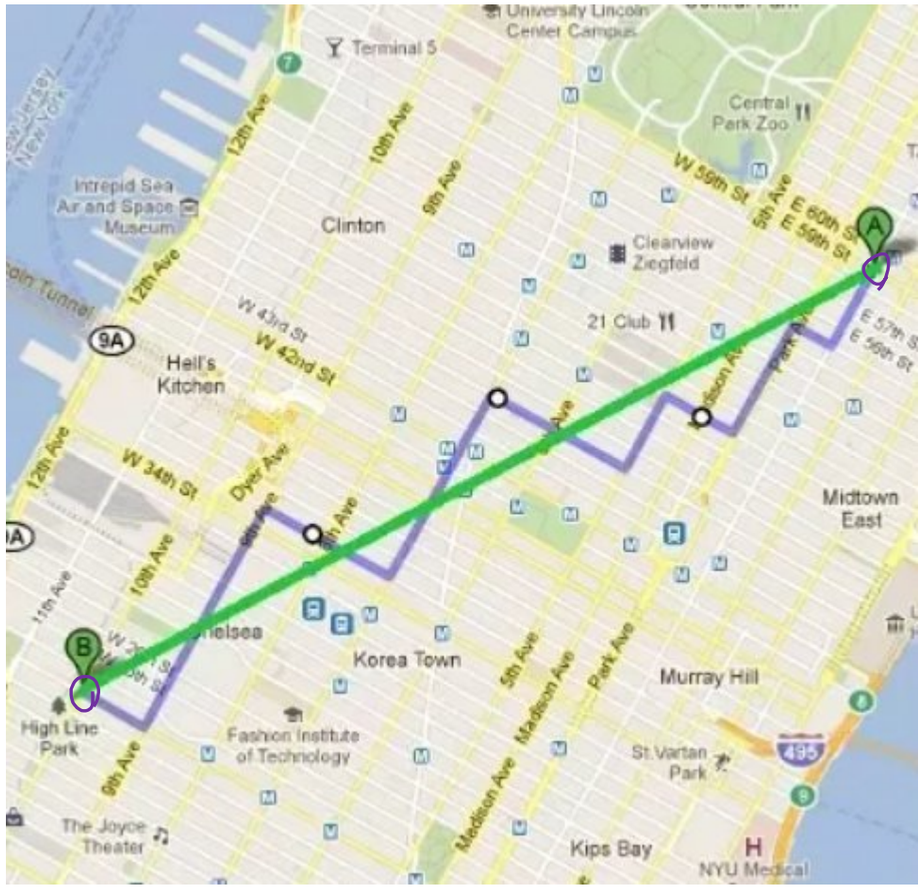
- Suppose two data points, both in R^d
  - $x^{(1)} = (x_1^{(1)}, x_2^{(1)}, \ldots, x_d^{(1)})$
  - $y^{(2)} = (y_1^{(2)}, y_2^{(2)}, \ldots, y_d^{(2)})$
- Euclidean distance: $d(x, y) = \sqrt{\sum_{i=1}^{d}(x_i - y_i)^2}$
- Minkowski distance: $d(x, y) = \sqrt[p]{\sum_{i=1}^{d}(x_i - y_i)^p}$
  - Euclidean distance: $p = 2$
  - Manhattan distance: $p = 1, d(x, y) = \sum_{i=1}^{d}|x_i - y_i|$
  - "inf"-distance: $p = \infty, d(x, y) = \max_{i=1}^{d}|x_i - y_i|$

Example



- Euclideandistance: $\sqrt{4^2 + 3^2} = 5$
- Manhattan distance: $4 + 3 = 7$
- “inf”-distance: $max\{4,3\} = 4$

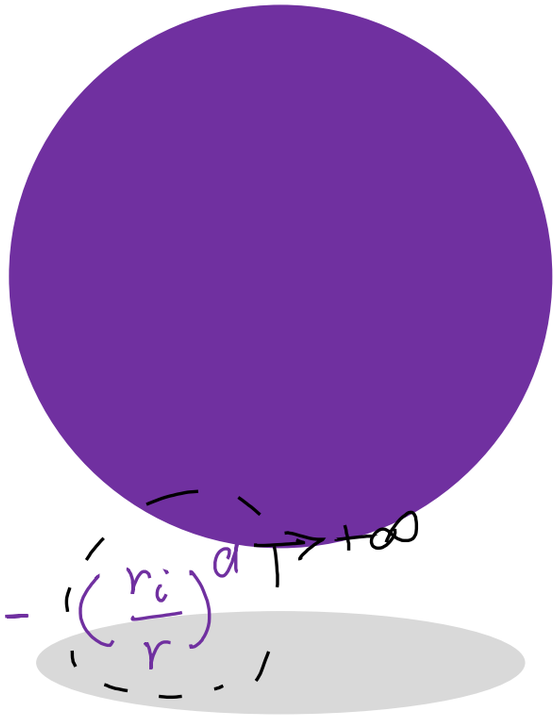
Some problems with Euclidean distance



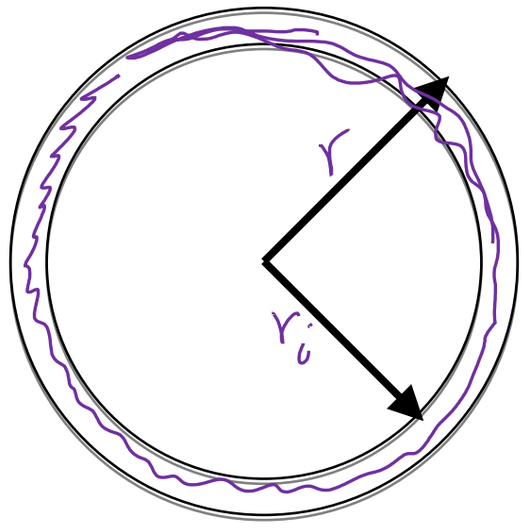
$$V_{total} = \frac{4}{3}\pi r^d = cr^d$$

$$V_{shell} = cr^d - cr_i^d$$

$$\frac{V_{shell}}{V_{total}} = \frac{cr^d - cr_i^d}{cr^d} =$$



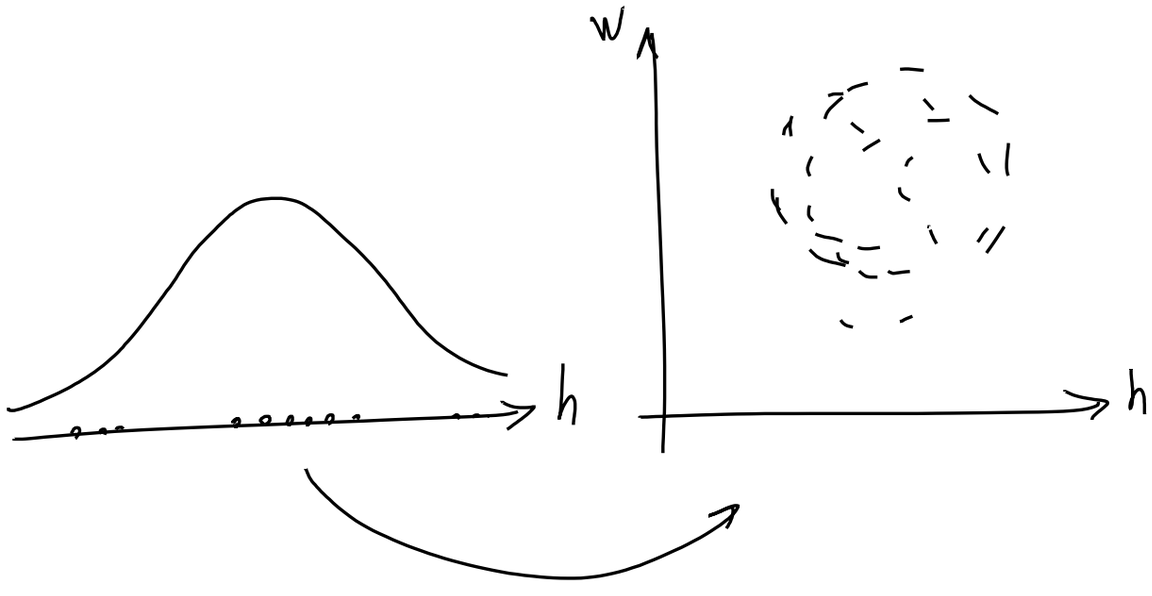
$$= 1 - \left(\frac{r_i}{r}\right)^d \quad d \to +\infty$$



$$r_i \leq r \Rightarrow \frac{r_i}{r} \leq 1$$

$$0 < \frac{r_i}{r_i} \leq 1$$

$$d \to +\infty \rightsquigarrow V_{shell} \approx V_{total}$$



Hamming Distance

- Manhattan distance is also called *Hamming distance* when all features are binary
  - Count the number of difference between two binary vectors
  - Example, $x, y \in \{0,1\}^{17}$

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
→ x	0	1	1	0	0	1	0	0	1	0	0	1	1	1	0	0	1
→ y	0	1	1	1	0	0	0	0	1	1	1	1	1	1	0	1	1

$$d(x, y) = 5$$

Edit Distance

- Transform one of the objects into the other, and measure how much effort it takes

x I N T E * N T I O N

| | | | | | | | | |

y * E X E C U T I O N

d s s i s

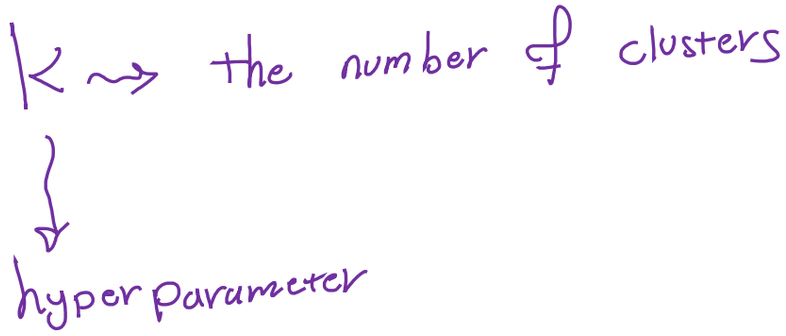
d: deletion (cost 5)
s: substitution (cost 1)
i: insertion (cost 2)

$$d(x, y) = 5 \times 1 + 3 \times 1 + 1 \times 2 = 10$$

Outline

- Clustering
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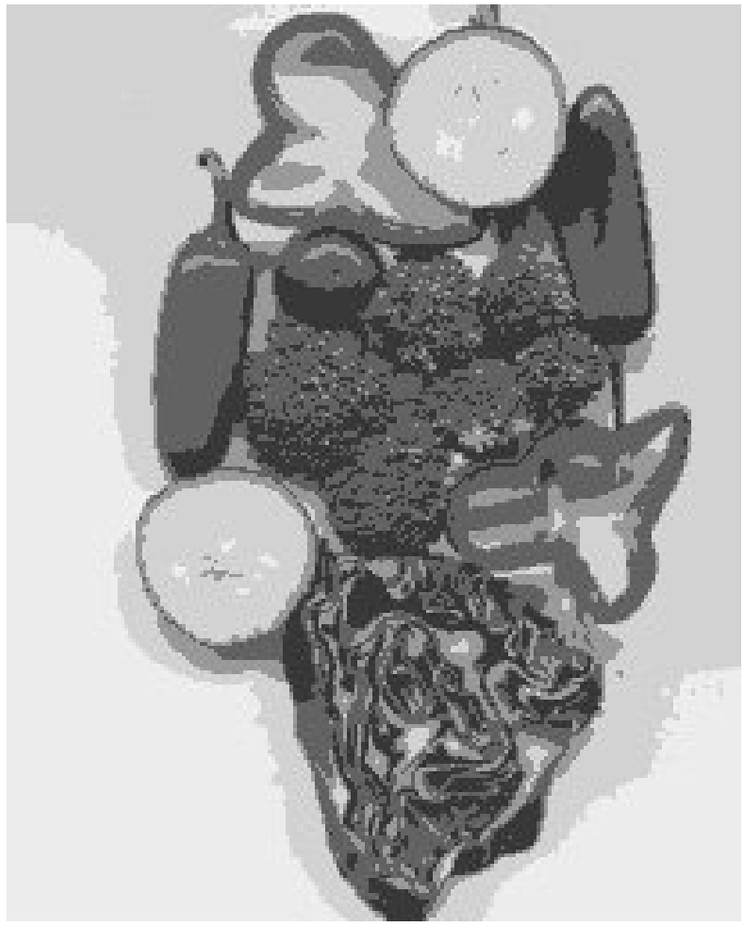




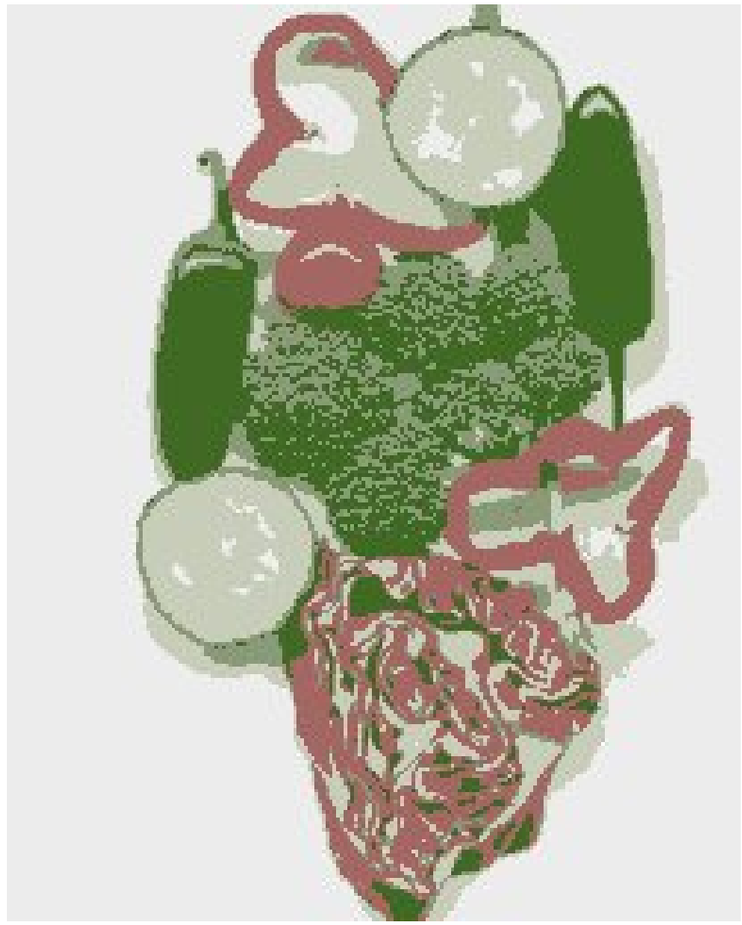
Results of K-Means Clustering:



Image



Clusters on intensity

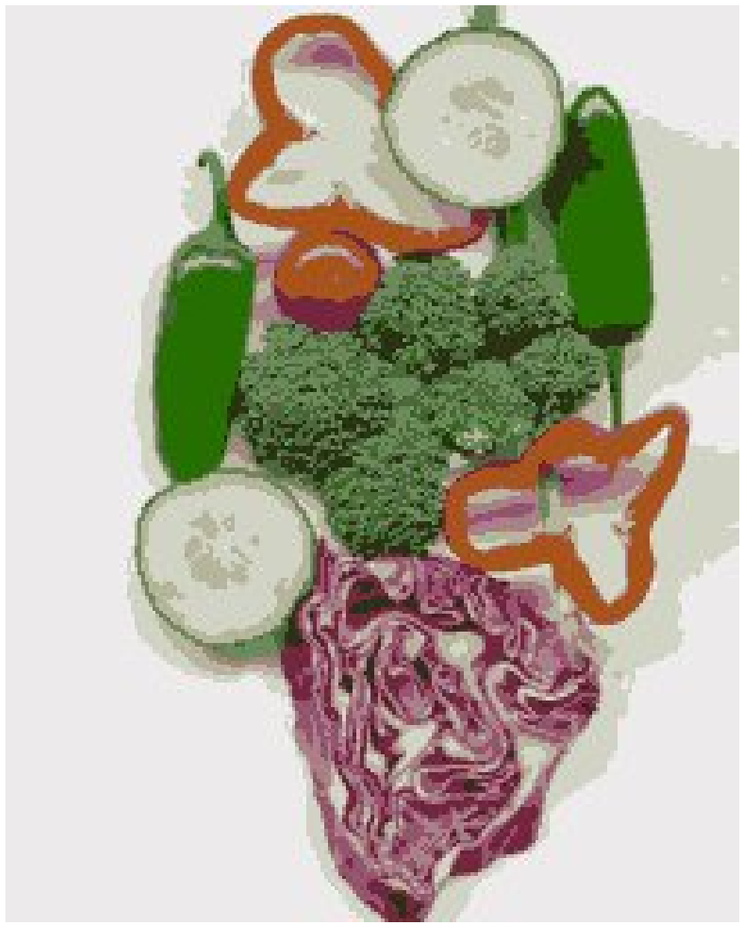


Clusters on color

K-means clustering using intensity alone and color alone



Image



Clusters on color

K-means using color alone, 11 segments (clusters)

1



2



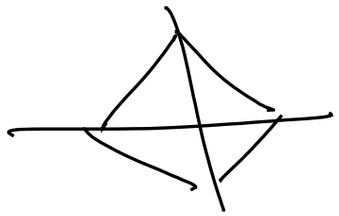


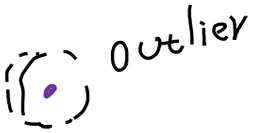


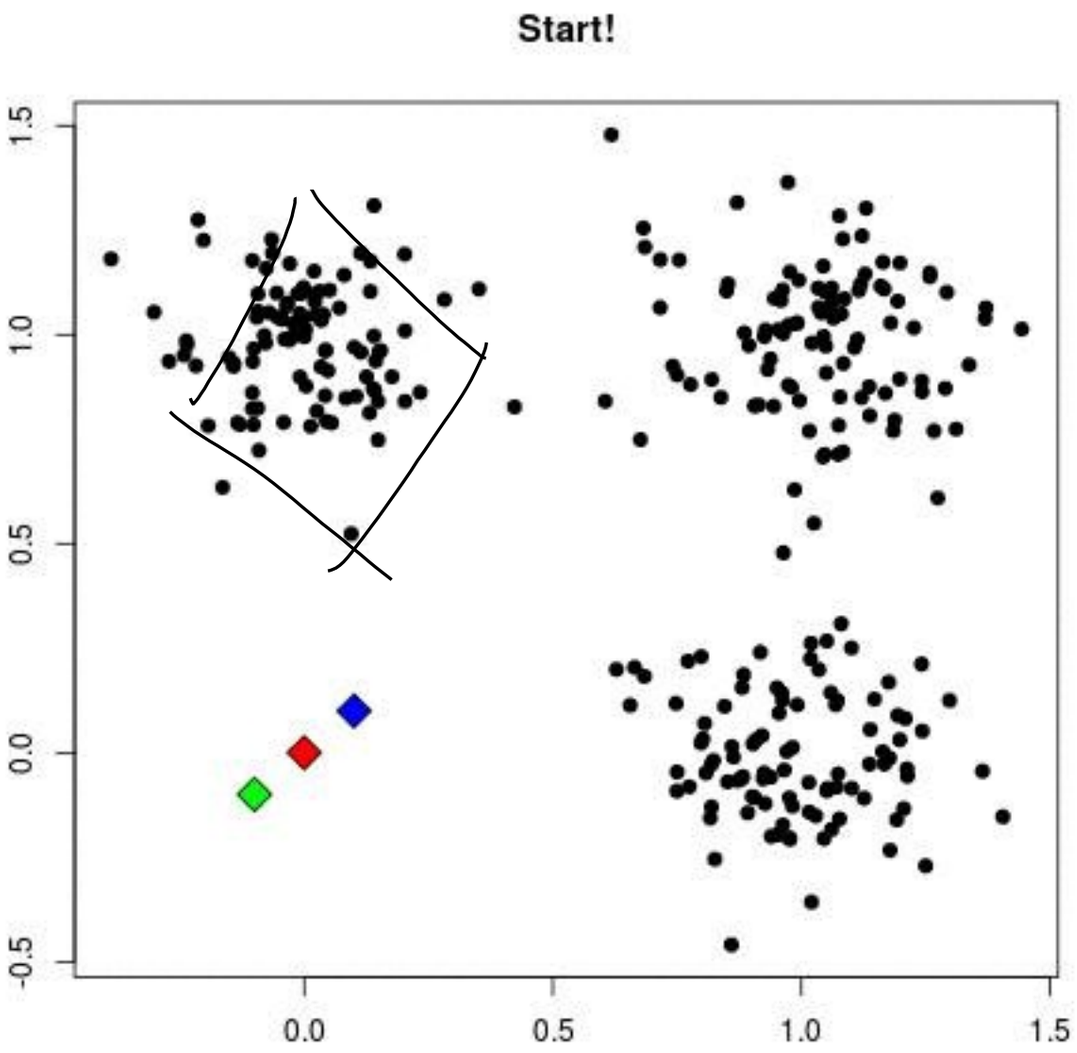
* Pictures from Mean Shift: A Robust Approach toward Feature Space Analysis, by D. Comaniciu and P. Meer

k - Medians

K-Means Algorithm







[Visualizing K-Means Clustering]

K-Means Algorithm

- Initialize k cluster centers, $\{c_1, c_2, \ldots, c_k\}$, randomly

- Do

Hard clustering

1. Decide the cluster memberships of each data point, x_i by assigning it to the nearest cluster center (cluster assignment)

$$\pi(i) = argmin_{j=1,\ldots,k} \left\| x_i^{(i)} - c_j^{(j)} \right\|^2$$

Expectation

2. Adjust the cluster centers (center adjustment)

EM

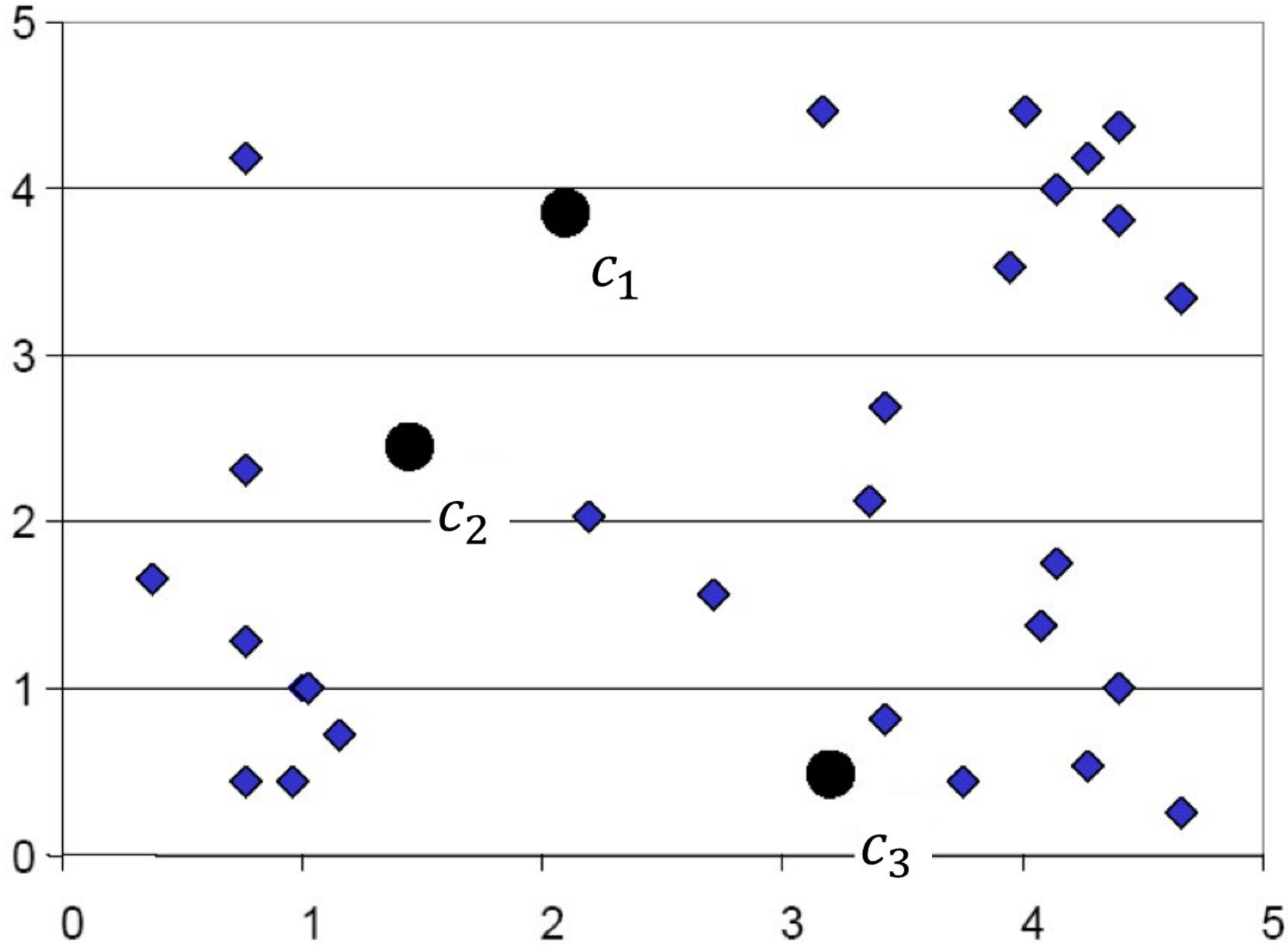
$$c_j = \frac{1}{|\{i : \pi(i) = j\}|} \sum_{i:\pi(i)} x_i^{(j)}$$

$$c_1 = \frac{1}{N_g} \sum x^{(i)}$$

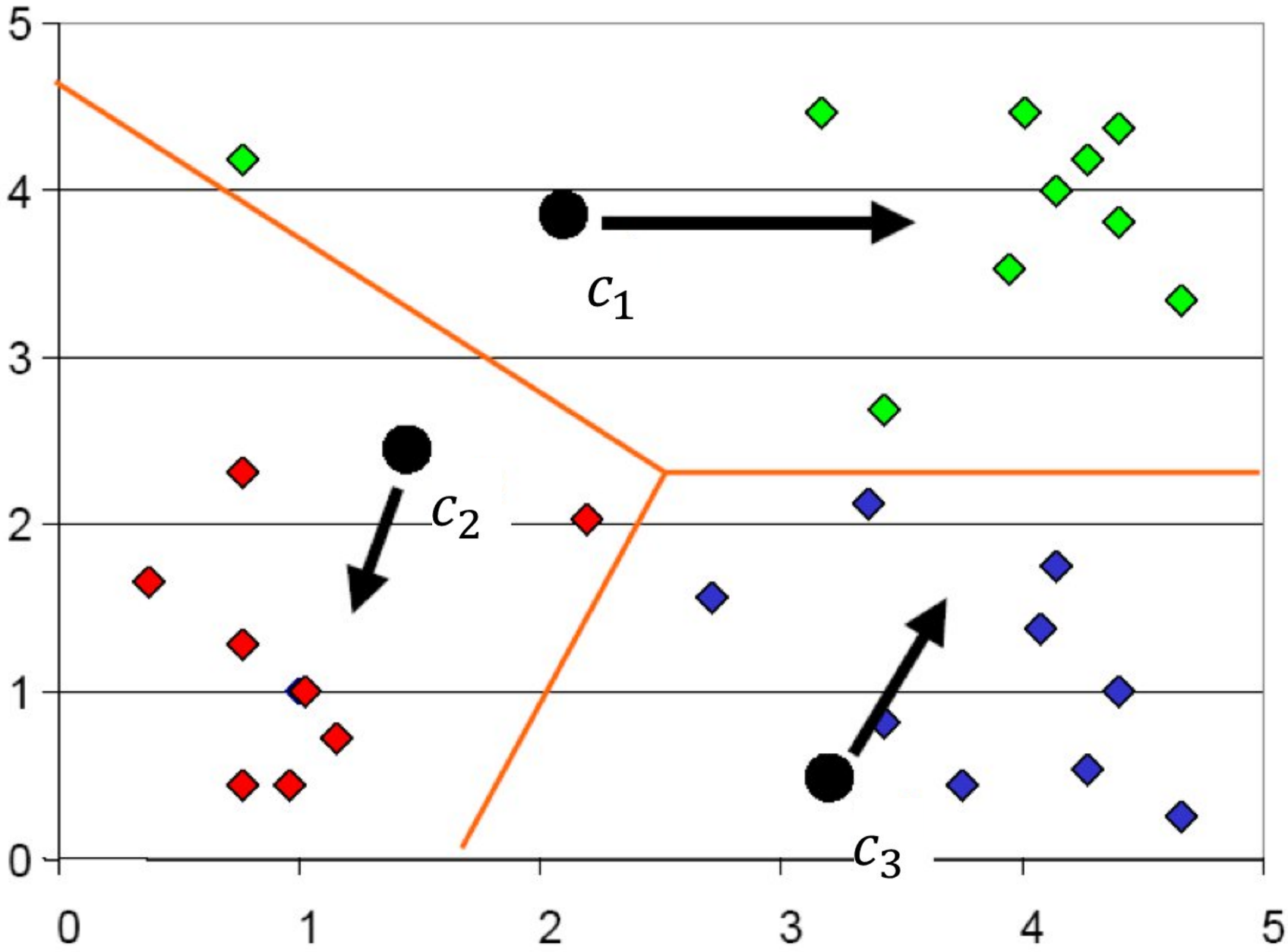
Maximization

- While any cluster center has been changed

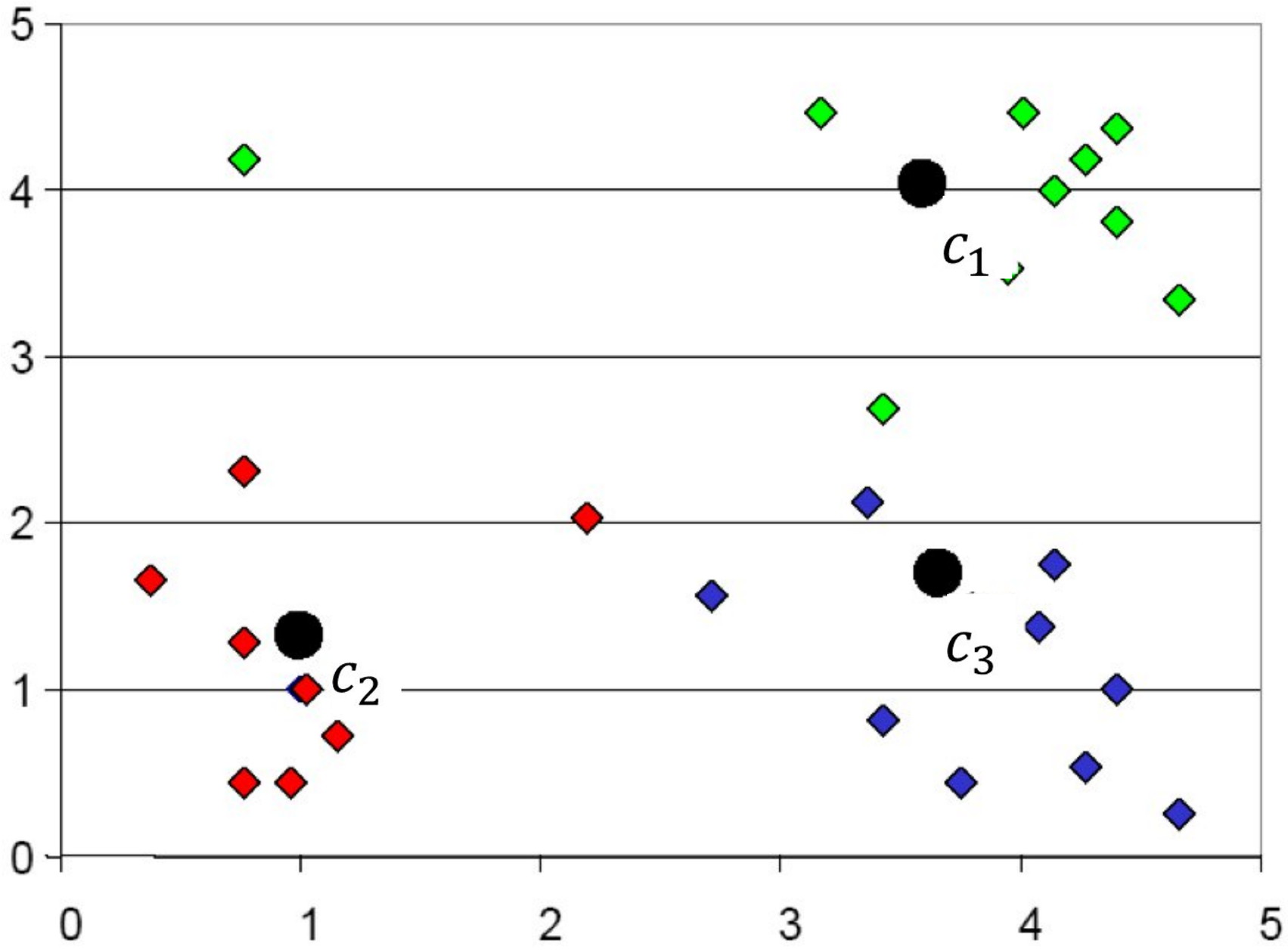
K-Means: Step 1



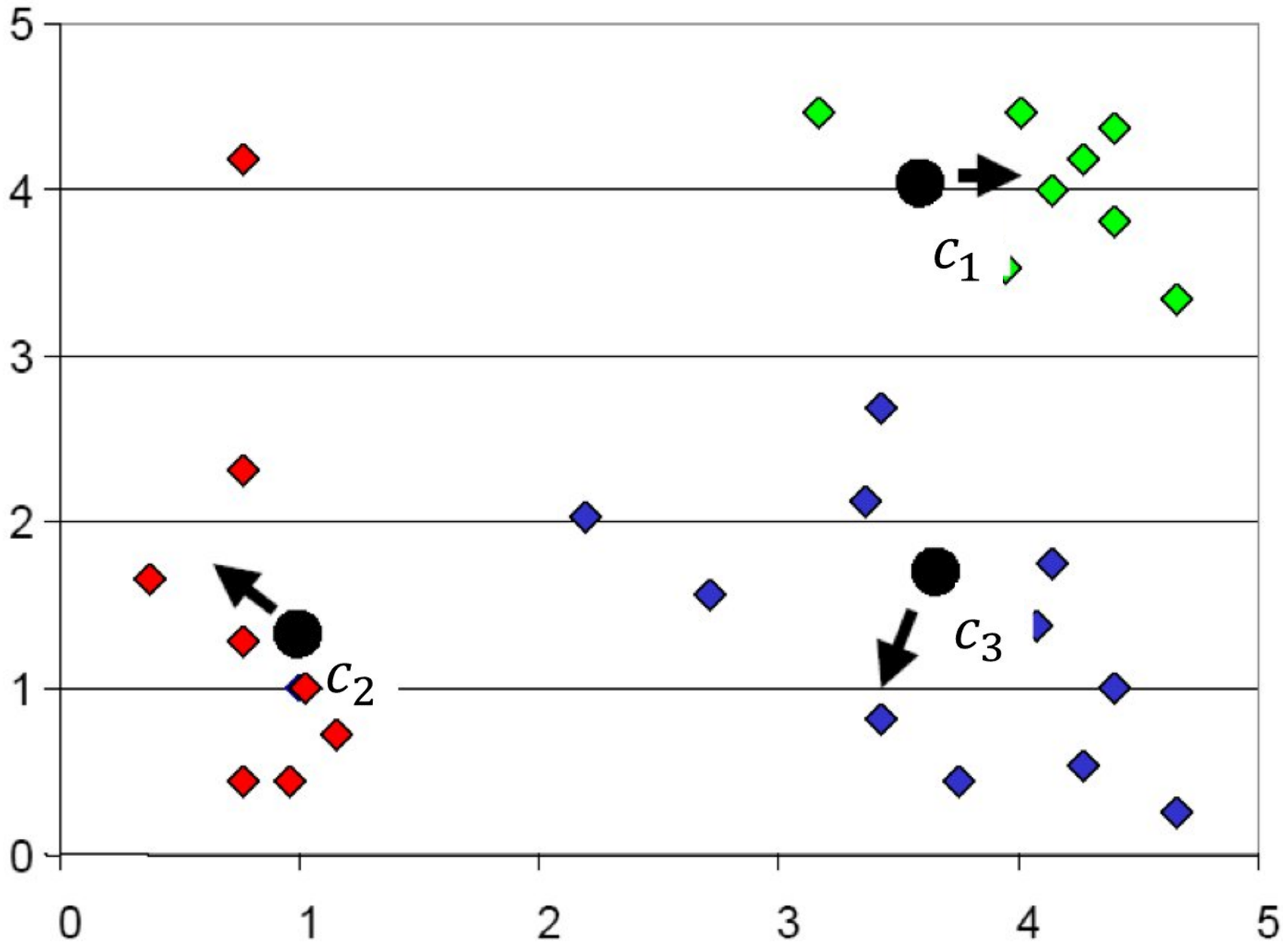
K-Means: Step 2



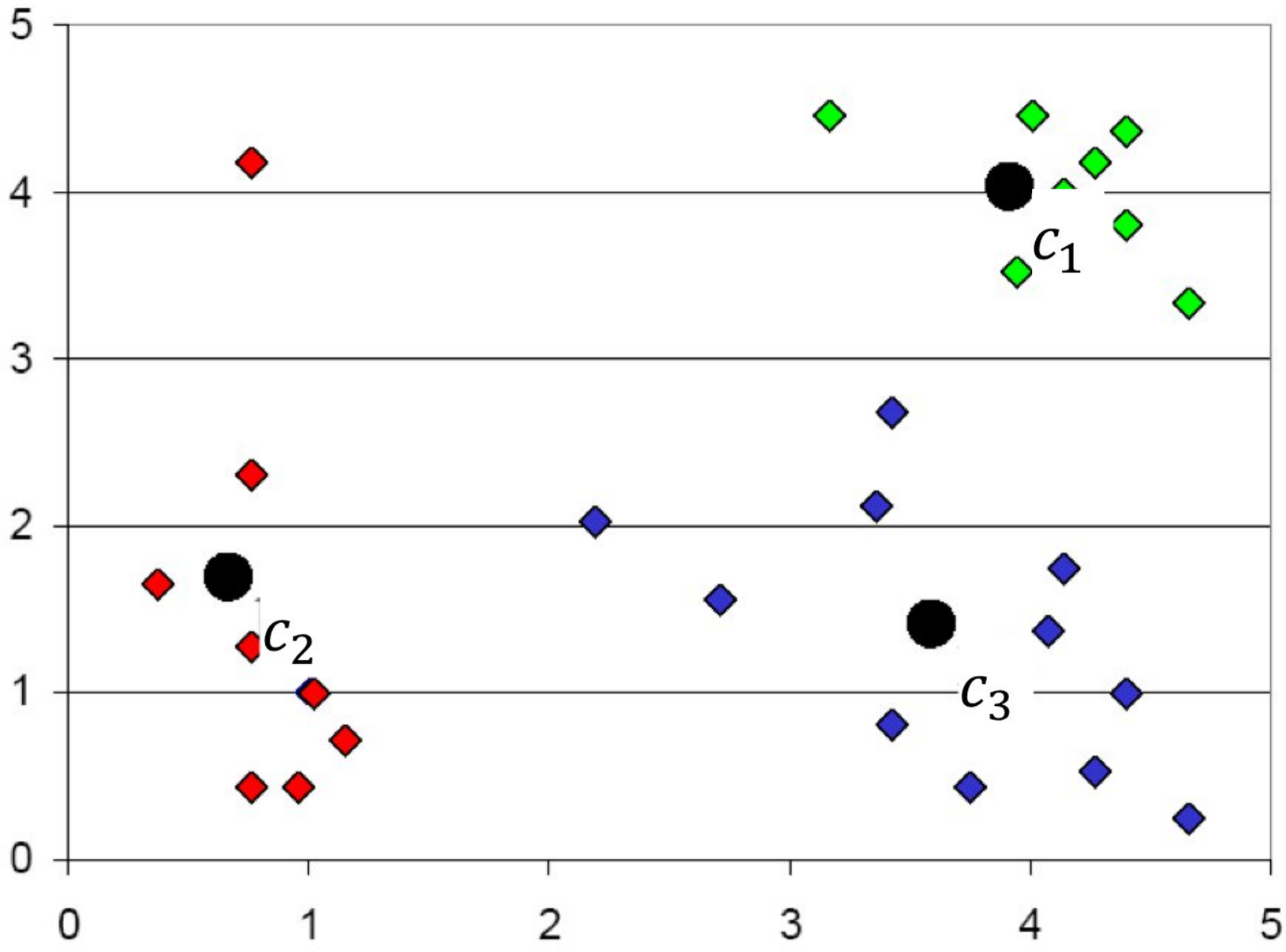
K-Means: Step 3



K-Means: Step 4



K-Means: Step 5

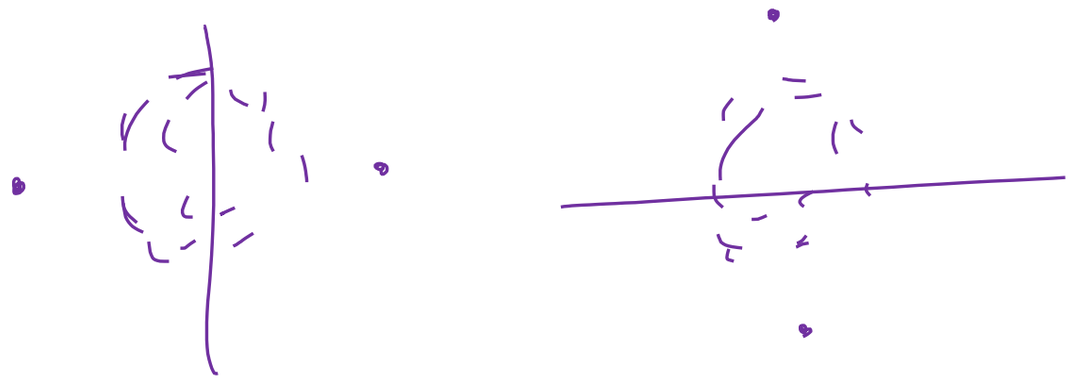


Outline

- Clustering
- Distance Function
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Questions

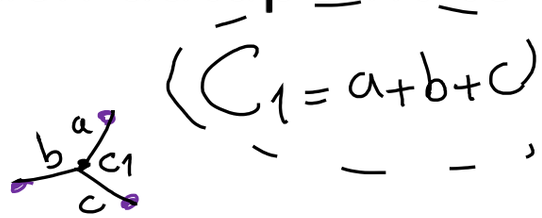
- Will different initialization lead to different results?
  - Yes
  - No
  - Sometimes

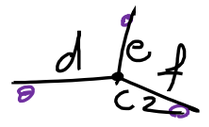


- Will the algorithm always stop after some iteration?
  - Yes
  - No (we have to set a maximum number of iterations)
  - Sometimes

Formal Statement of the Clustering Problem

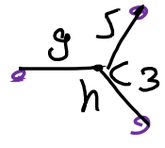
- Given n data points, $\{x_1, x_2, \dots, x_n\}\ x \in R^d$
- Find k cluster centers, $\{c_1, c_2, \dots, c_k\}\ c \in R^d$
- And assign each datapoint i to one cluster, $\pi(i) \in \{1, \dots, k\}$
- Such that the averaged square distances from each datapoint to its respective cluster center is small





$C_2 = d + e + f$

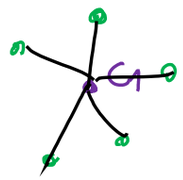
$$\text{distortion} = \min_{c,\pi} \sum_{i=1}^{n} \left\| x^{(i)} - c_{\pi(i)} \right\|^2$$



$C_3 = g + h + i$

GIR

$\text{distortion} = C_1 + C_2 + C_3$



$$D = \min \sum_{i=1}^{Ng} \| x^{(i)} - c_g \|^2$$

$$= \min \left(\sum_{i=1}^{Ng} x^{(i)2} - 2 \sum_{i=1}^{Ng} x^{(i)} c_g + \left(\sum_{i=1}^{Ng} c_g^2 \right) \right)$$

$$N_g c_g^2$$

$$\frac{\partial D}{\partial c_g} = 0 \Rightarrow -2 \sum_{i=1}^{Ng} x^{(i)} + 2 N_g c_g = 0$$

$$-2 \sum_{i=1}^{Ng} x^{(i)} = -2 N_g c_g \Rightarrow \boxed{c_g = \frac{1}{N_g} \sum_{i=1}^{N} x^{(i)}}$$

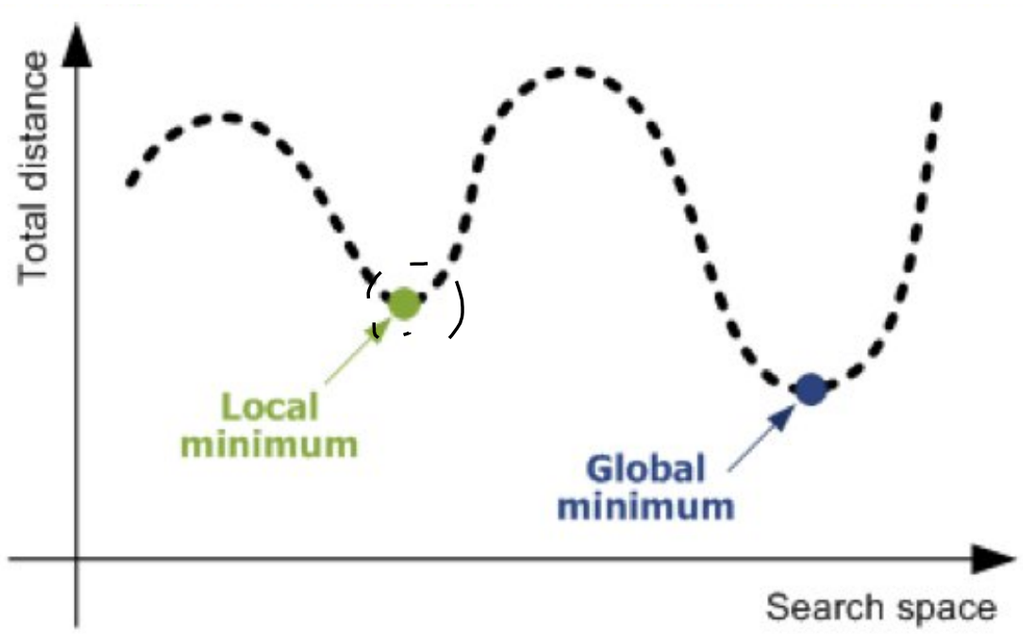
Clustering is NP-Hard

- Find k cluster centers, $\{c_1, c_2, \ldots, c_k\}$ c $\in R^{\text{d}}$, and assign each data point i to one cluster, $\pi(i) \in \{1, \ldots, k\}$, to minimize

$$\min_{c,\pi} \sum_{i=1}^{n} \left\| x_i - c_{\pi(i)} \right\|^2$$



- A search problem over the space of discrete assignments
  - For all n data point together, there are k^{n} possibility
  - The cluster assignment determines cluster centers, and vice versa



- For all n data point together, there are k^n possibility

$k^n = 2^3 = 8$

X = {A,B,C}

n=3 (data points)

k=2 clusters of two members

Cluster 1	Cluster 2
A, B, C	{ }
{ }	A, B, C
A	B, C
B	A, C
C	A, B
B, C	A
A, C	B
A, B	C

Convergence of K-Means

- Will kmeans objective oscillate?

$$\min_{c,\pi} \sum_{i=1}^{n} \left\| x_i - c_{\pi(i)} \right\|^2$$

- The minimum value of the objective is finite

- Each iteration of kmeans algorithm decrease the objective
  - Cluster assignment step decreases objective
    - $\pi(i) = argmin_{j=1,\ldots,k} \ \left\| x_i - c_{\pi(j)} \right\|^2$ for each data point i
  - Center adjustment step decreases objective
    - $\boldsymbol{c_i} = \frac{1}{|\{i:\pi(i)=j\}|} \sum_{i:\pi(i)=j} \boldsymbol{x_i} = \boldsymbol{argmin_c} \sum_{i:\pi(i)=j} \left\| \boldsymbol{x_i} - \boldsymbol{c_{\pi(j)}} \right\|^2$

Time Complexity

- Assume computing distance between two instances is O(d) where d is the dimensionality of the vectors.

$$\|x^{(1)} - x^{(2)}\|^2 = \underbrace{(x_1^{(1)} - x_1^{(2)})^2}_{O(d)} + (x_2^{(1)} - x_2^{(2)})^2 + \cdots + (x_d^{(1)} - x_d^{(2)})^2$$

- Reassigning clusters for all datapoints:
  - O(kn) distance computations (when there is one feature)
  - O(knd) (when there is d features)

$O(nd)$

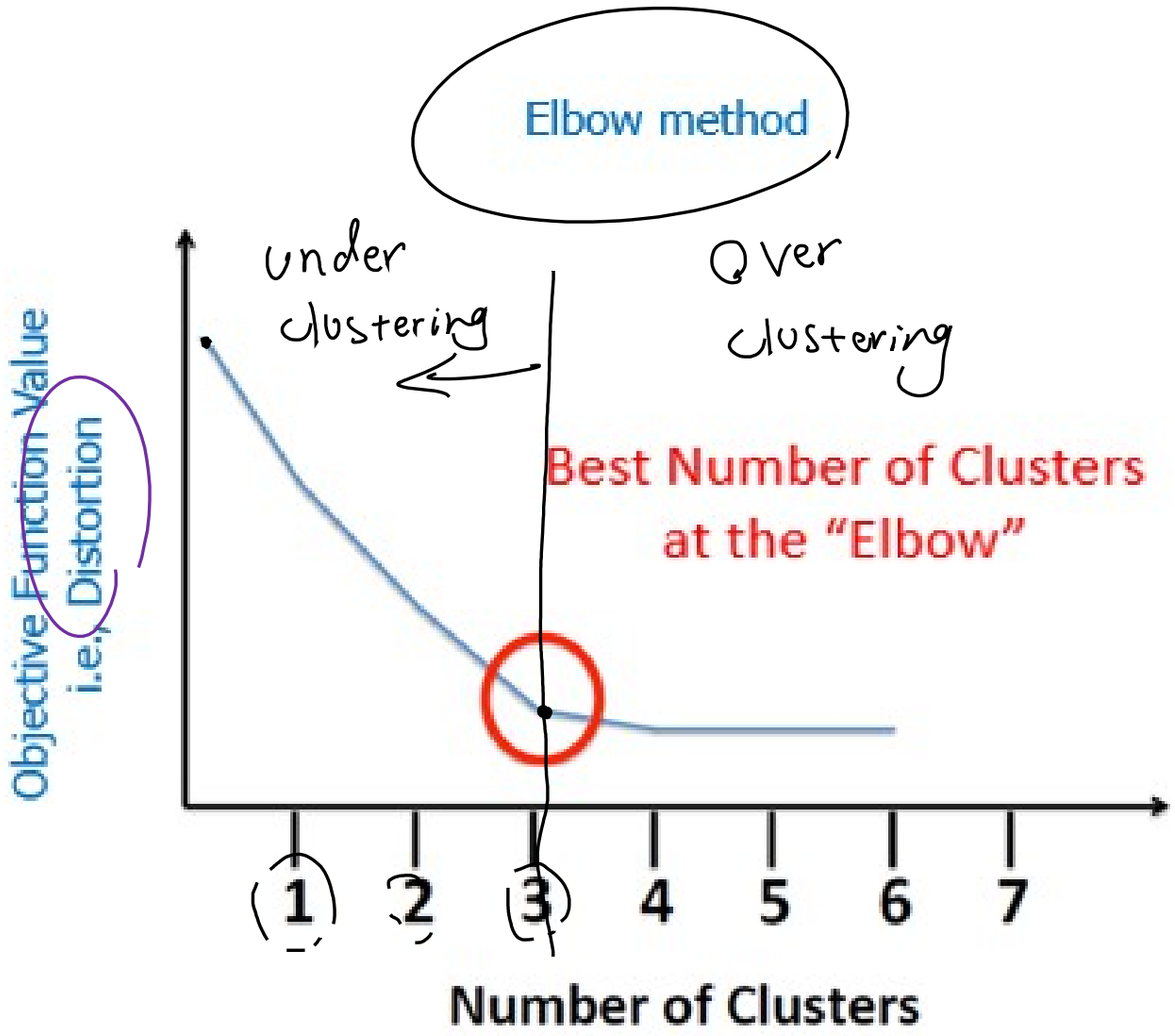
$O(knd)$

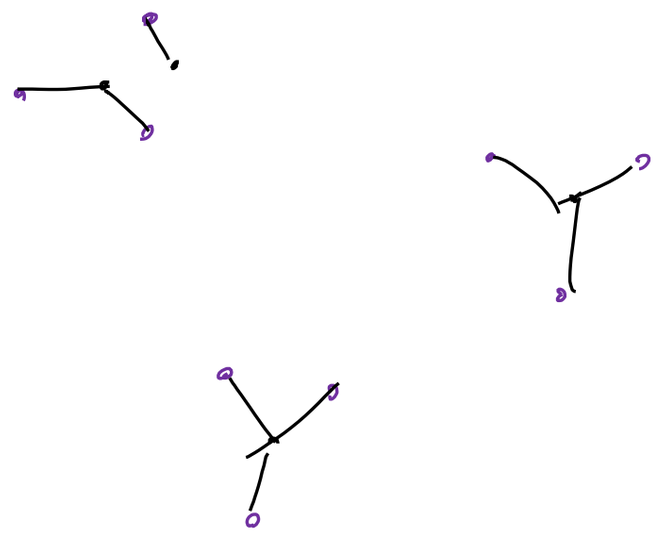
- Computing centroids: Each instance vector gets added once to some centroid (Finding centroid for each feature): O(nd).

- Assume these two steps are each done once for I iterations: O(Iknd).

$O(Iknd)$

How to Choose K?





Drawbacks of k-means:

1. Sensitive to outliers
2. It works the best for spherical-shaped clusters
3. We need to know k in advance

Distortion score: computing the sum of squared distances from each point to its assigned center

Image credit: Dileka Madushan.